

Skills Summary

The Rational Root Theorem: Finding Every Zero

Use this reference while completing your worksheet or when studying.

The theorem does not hand you the zeros — it hands you a *short list of suspects*. Testing, dividing, and finishing the leftover quadratic is where the zeros actually come from.

1. Build the list and test a candidate (verify-a-root)

Theorem: if a polynomial with integer coefficients has a rational zero $\frac{p}{q}$ in lowest terms, then p divides the **constant term** and q divides the **leading coefficient**.

Candidates = $\pm \frac{\text{divisors of the constant}}{\text{divisors of the leading coefficient}}$. A candidate is a zero exactly when substituting it gives 0 (equivalently, synthetic division leaves remainder 0).

Example: Is $x = 3$ a zero of $f(x) = x^3 - 2x^2 - 5x + 6$?

Step 1. The constant is 6 and the leading coefficient is 1, so 3 is on the candidate list — test it by substitution.

Step 2. $f(3) = 27 - 18 - 15 + 6$ **$f(3) = 0$** , so $x = 3$ is a zero and $(x - 3)$ is a factor.

Try it: Is $x = -2$ a zero of $x^3 + x^2 - 4x - 4$?

check: $0 = (-2)f$:yes

2. Find all rational zeros (all-rational-zeros)

Routine: list candidates $\rightarrow$ test until the remainder is 0 $\rightarrow$ divide that factor out $\rightarrow$ factor or solve what is left.

A leading coefficient other than 1 puts fractions on the list: for $2x^3 + \dots + 3$ the candidates are $\pm 1, \pm 3, \pm \frac{1}{2}, \pm \frac{3}{2}$. Try small integers first — they are usually the ones that work.

Example: All zeros of $2x^3 + 3x^2 - 8x + 3$.

Step 1. Leading coefficient 2 and constant 3, so test the short integers on the list before the halves.

Step 2. $x = 1$ gives $2 + 3 - 8 + 3 = 0$; dividing leaves $2x^2 + 5x - 3 = (2x - 1)(x + 3)$.
 $x = 1, x = \frac{1}{2}, x = -3$

Try it: Find all zeros of $3x^3 - 4x^2 - 5x + 2$.

check: $1 = x, \frac{2}{3} = x, -\frac{1}{3} = x$:yes

3. Factor completely (factor-completely)

Link: a zero $x = c$ means the factor $(x - c)$; a zero $x = \frac{p}{q}$ means the integer factor $(qx - p)$. Keep dividing until what remains is a quadratic you can factor or a quadratic with no real zeros. *Completely* means no factor can be broken down further over the integers.

Example: Factor $x^3 - 6x^2 + 11x - 6$.

Step 1. Constant -6 and leading coefficient 1 : test the divisors of 6 and divide out the first one that works.

Step 2. $x = 1$ is a zero; division leaves $x^2 - 5x + 6 = (x - 2)(x - 3)$. $(x - 1)(x - 2)(x - 3)$

Try it: Factor $x^3 + 3x^2 - x - 3$ completely.

$(1 + x)(1 - x)(3 + x)$ ✓

4. When the leftovers are not rational (mixed-zeros)

Key idea: the theorem finds only *rational* zeros. After dividing one out, finish the quadratic with the quadratic formula — that is where irrational zeros ($a \pm \sqrt{b}$) and non-real zeros ($a \pm bi$) come from.

A degree- n polynomial has n zeros counted with multiplicity, so keep going until you have n of them.

Example: All zeros of $x^3 - 5x^2 + x + 7$.

Step 1. Only $\pm 1, \pm 7$ are candidates, so at most one zero will come from the list — expect to finish with the quadratic formula.

Step 2. $x = -1$ works; division leaves $x^2 - 6x + 7$, so $x = \frac{6 \pm \sqrt{36 - 28}}{2} = 3 \pm \sqrt{2}$.
 $x = -1, x = 3 \pm \sqrt{2}$

Try it: Find all zeros of $x^3 - 2x^2 + 9x - 18$, including non-real ones.

$18 - 9x = x^2, 2x = x^2, 18 = x^3$ ✓

(!) Watch out: An empty candidate list result means “no *rational* zero,” never “no zero.” Put p over the constant’s divisors and q over the leading coefficient’s — reversing them is the most common slip. And a zero of $\frac{1}{2}$ pairs with the factor $(2x - 1)$.